

Predicting Marathon Race Performance Using Machine Learning

Project Progress Update #1

IECE 465: Introduction to Machine Learning for Engineers

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AGENDA

1. Project Overview
2. Dataset Description
3. Data Preprocessing
4. Model Implementation
5. Preliminary Results
6. Live Demonstration
7. Future Work



PROJECT MOTIVATION

Q: Why Predict Marathon Finish Times?

Marathon performance depends on many factors including:

- Age
- Gender
- Pacing strategy
- Physical endurance

Goal of this Project:

- Predict a runner's **net marathon finish time**
- Compare performance of:
 1. Linear Regression (baseline)
 2. Neural Network Regression

Applications:

- Sports analytics
- Training optimization
- Performance prediction



DATASET OVERVIEW

Dataset: 2023 Boston Marathon

Source: Boston Marathon Results Dataset

Dataset Size:

- 26,598 runners
- 15 total variables

Key Features Used:

- *age_group*
- *gender*
- *half_time_sec*
- *finish_net_sec* (target variable)

Example Variables Available:

- Overall placement
- Gender placement
- Division placement
- Half-marathon split time
- Gun time vs. net time



DATASET PREPROCESSING

Data Cleaning Steps

1. Imported dataset into MATLAB:

```
data = readtable('boston_marathon_2023.csv');
```

2. Selected relevant features

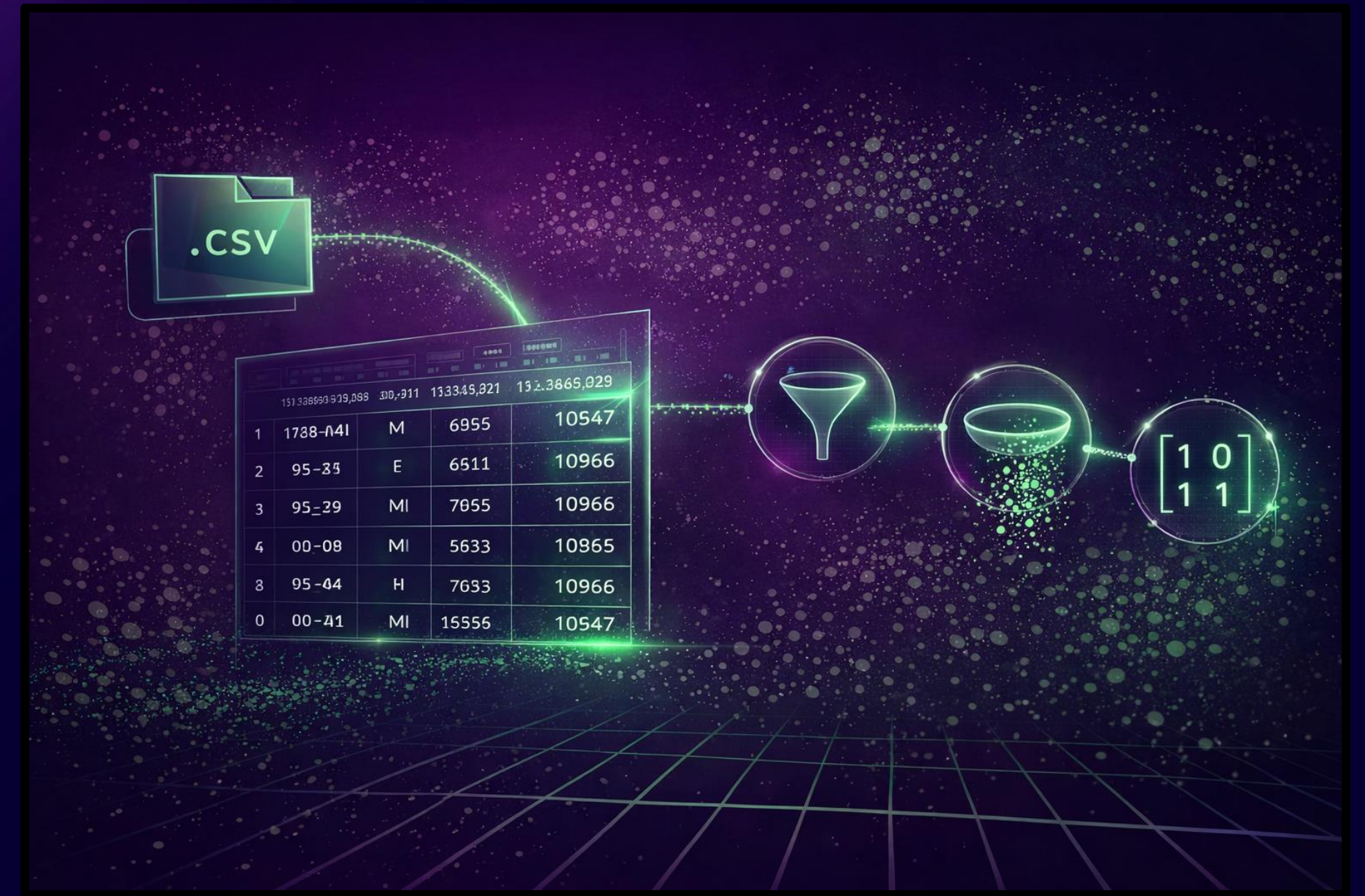
3. Removed missing values:

```
data = rmmissing(data);
```

4. Converted categorical variables:

```
data.age_group = categorical(data.age_group);
```

```
data.gender = categorical(data.gender);
```



DATA VISUALIZATION

Understanding the Dataset

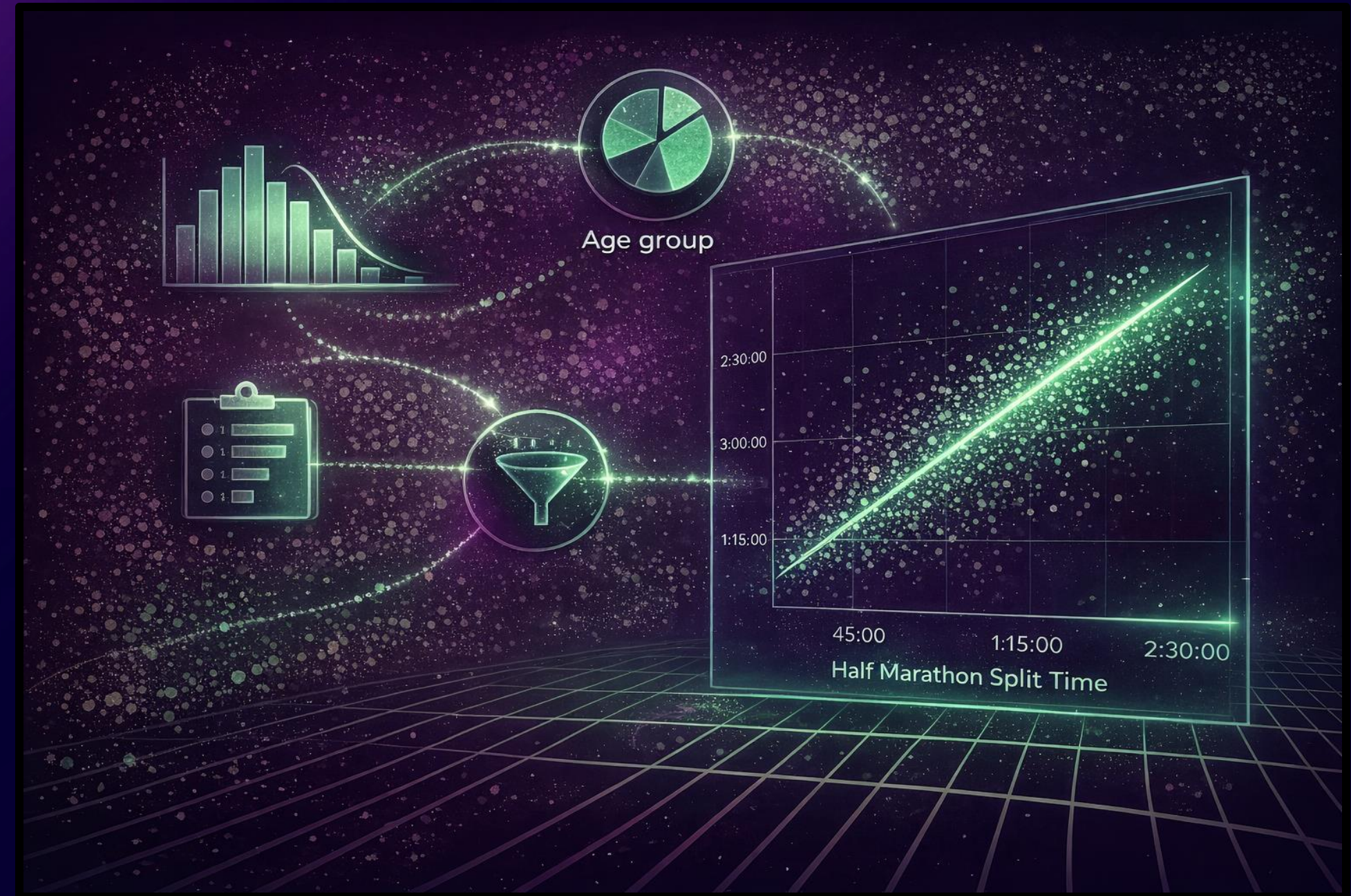
Possible Visualizations:

- Distribution of finish times
- Half marathon split vs final finish time
- Age group distribution

Key insight: Half-marathon split time is strongly correlated with final marathon time.

Suggested graph to include:

- Scatter plot
 - X-axis: half marathon time
 - Y-axis: final finish time



MODEL 1: LINEAR REGRESSION

Baseline Model

We first trained a linear regression model.

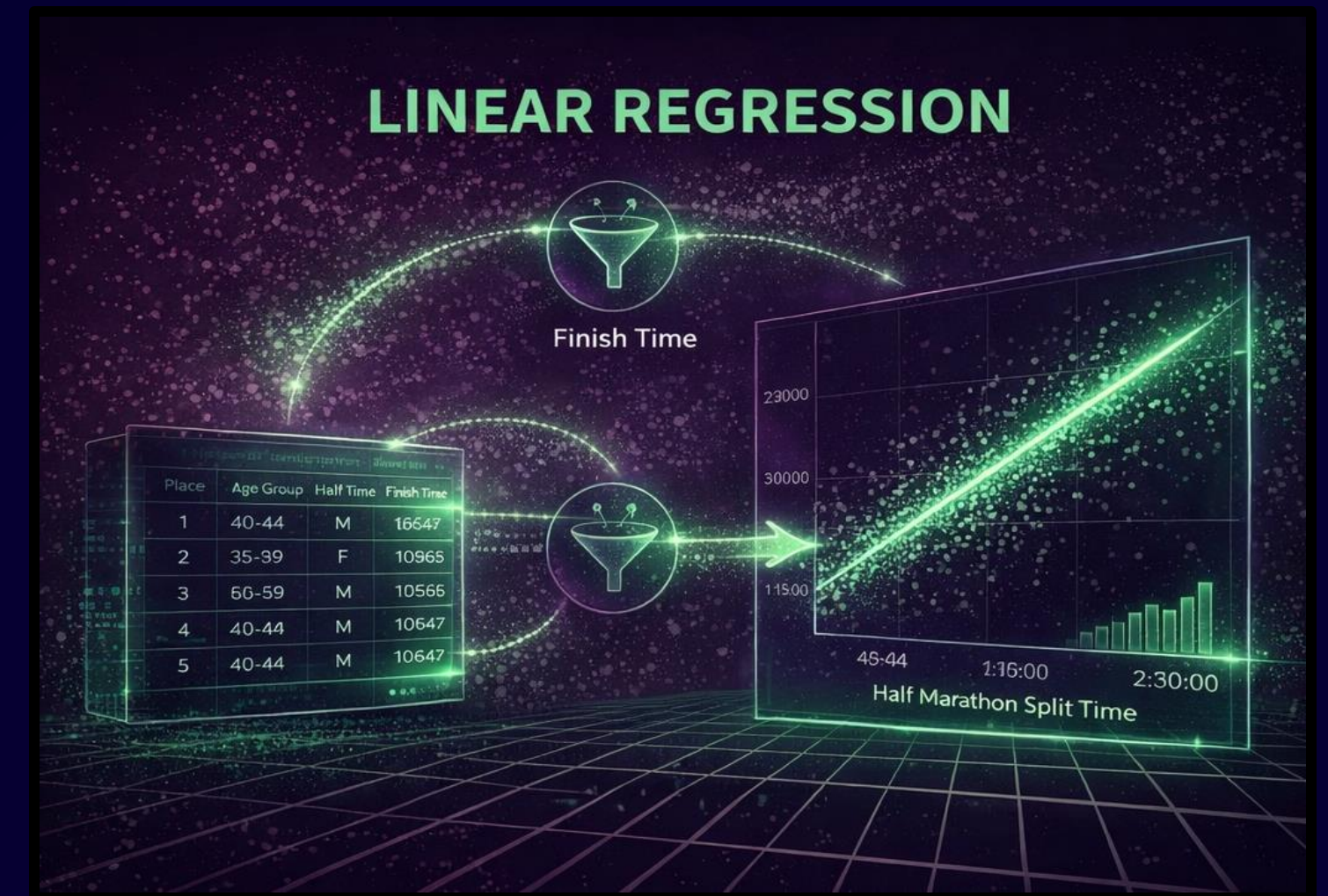
MATLAB Implementation:

```
linModel = fitlm(trainData, 'finish_net_sec ~ age_group + gender + half_time_sec');
```

Prediction: `y_pred_lin = predict(linModel, testData);`

Evaluation metrics:

- Mean absolute error (MAE)
- Root Mean Squared Error (RMSE)



MODEL 2: NEURAL NETWORK

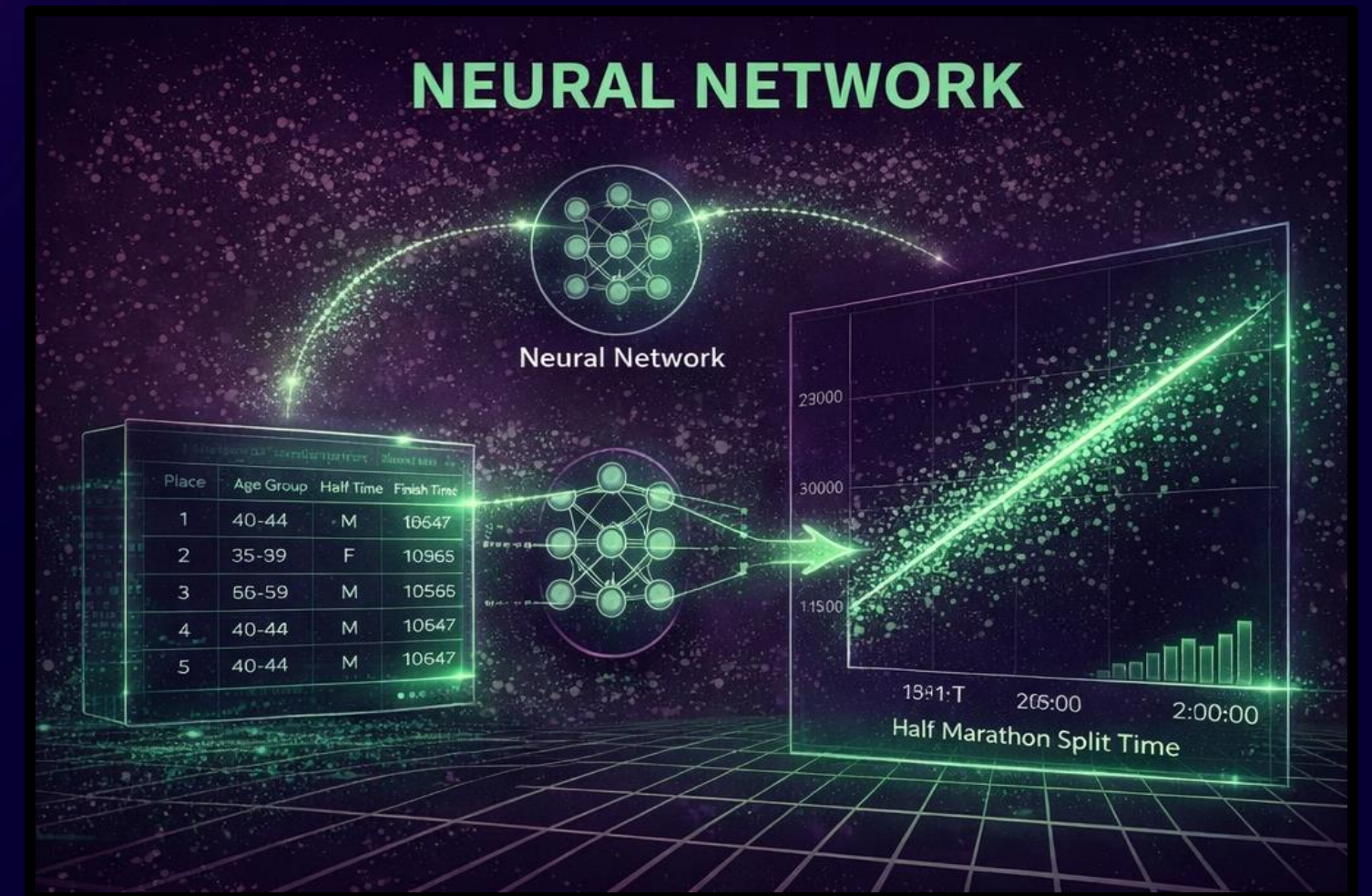
Neural Network Regression

Goal: Capture nonlinear relationships between features.

Model Architecture:

- Input features
- Hidden layers: 32 → 16 neurons
- Activation function: ReLU

MATLAB Implementation: `nnModel = fitrnet(Xtrain, ytrain, 'LayerSizes',[32 16], 'Activations','relu');`



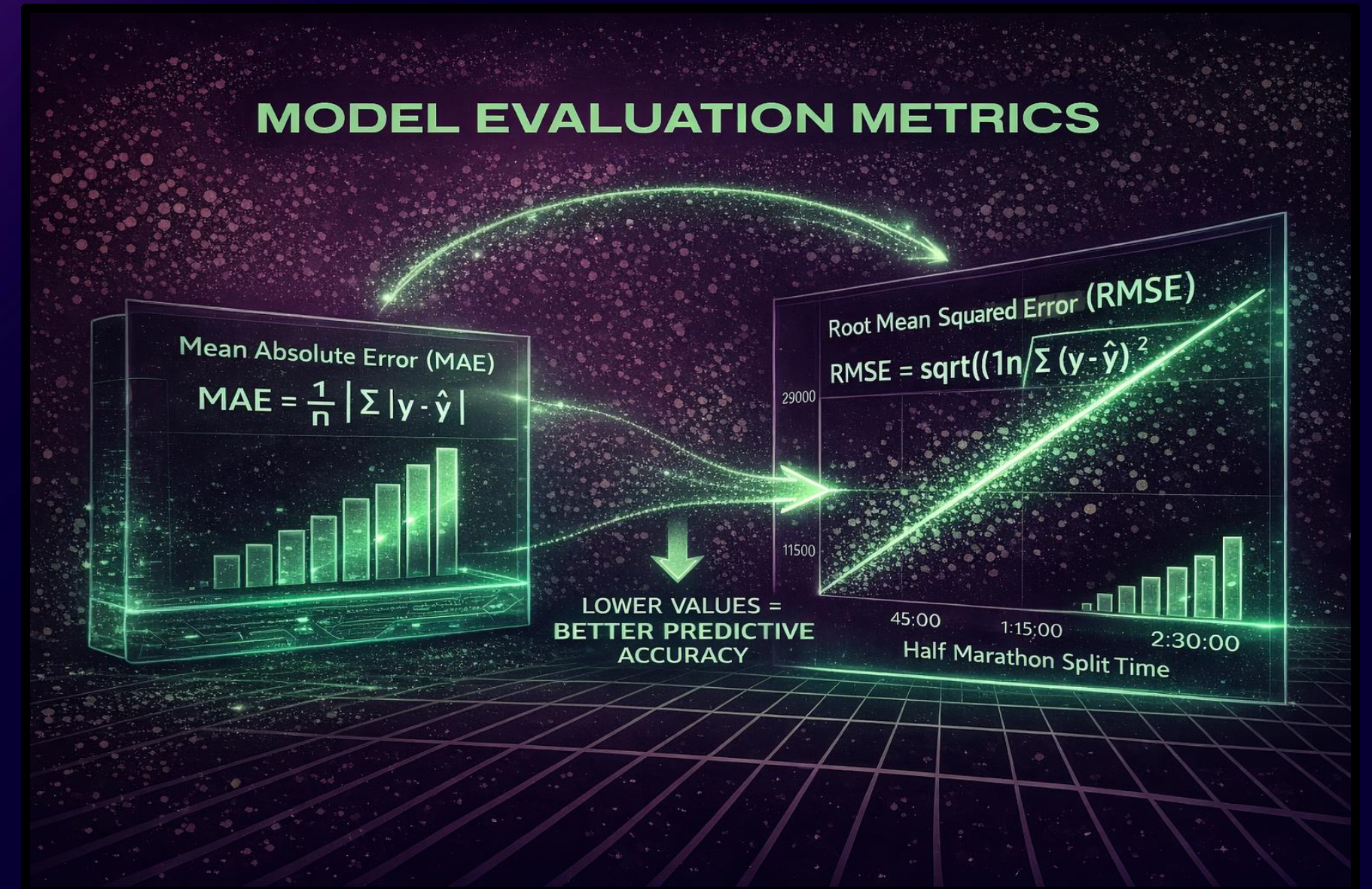
MODEL EVALUATION METRICS

Evaluation Metrics

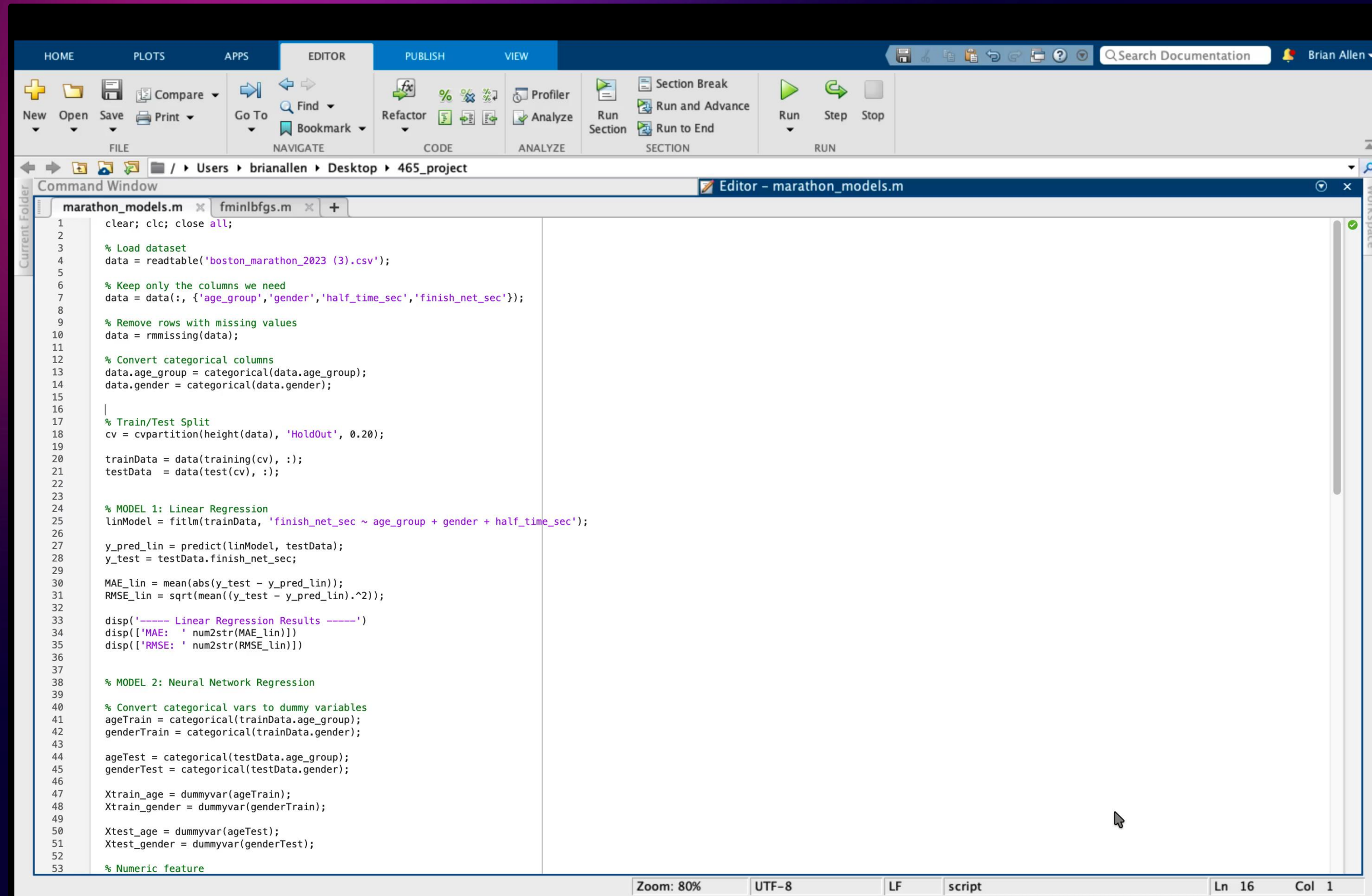
We evaluated model performance using:

- Mean Absolute Error (MAE):
$$\text{MAE} = (1/n) * \sum |y - \hat{y}|$$
- Root Mean Squared Error (RMSE):
$$\text{RMSE} = \text{sqrt}((1/n) * \sum (y - \hat{y})^2)$$

Lower values indicate better predictive accuracy.



SYSTEM DEMONSTRATION



The screenshot displays the MATLAB R2023a Editor interface. The top menu bar includes HOME, PLOTS, APPS, EDITOR, PUBLISH, and VIEW. The EDITOR tab is active, showing a toolbar with icons for file operations (New, Open, Save, Print), navigation (Go To, Find, Bookmark), code editing (Refactor), analysis (Profiler, Analyze), and execution (Run, Step, Stop). The Command Window shows the current folder path: Users > brianallen > Desktop > 465_project. The Editor window displays a script named 'marathon_models.m' with the following code:

```
1 clear; clc; close all;
2
3 % Load dataset
4 data = readtable('boston_marathon_2023 (3).csv');
5
6 % Keep only the columns we need
7 data = data(:, {'age_group', 'gender', 'half_time_sec', 'finish_net_sec'});
8
9 % Remove rows with missing values
10 data = rmmissing(data);
11
12 % Convert categorical columns
13 data.age_group = categorical(data.age_group);
14 data.gender = categorical(data.gender);
15
16 |
17 % Train/Test Split
18 cv = cvpartition(height(data), 'HoldOut', 0.20);
19
20 trainData = data(training(cv), :);
21 testData = data(test(cv), :);
22
23
24 % MODEL 1: Linear Regression
25 linModel = fitlm(trainData, 'finish_net_sec ~ age_group + gender + half_time_sec');
26
27 y_pred_lin = predict(linModel, testData);
28 y_test = testData.finish_net_sec;
29
30 MAE_lin = mean(abs(y_test - y_pred_lin));
31 RMSE_lin = sqrt(mean((y_test - y_pred_lin).^2));
32
33 disp('----- Linear Regression Results -----')
34 disp(['MAE: ' num2str(MAE_lin)])
35 disp(['RMSE: ' num2str(RMSE_lin)])
36
37
38 % MODEL 2: Neural Network Regression
39
40 % Convert categorical vars to dummy variables
41 ageTrain = categorical(trainData.age_group);
42 genderTrain = categorical(trainData.gender);
43
44 ageTest = categorical(testData.age_group);
45 genderTest = categorical(testData.gender);
46
47 Xtrain_age = dummyvar(ageTrain);
48 Xtrain_gender = dummyvar(genderTrain);
49
50 Xtest_age = dummyvar(ageTest);
51 Xtest_gender = dummyvar(genderTest);
52
53 % Numeric feature
```

The status bar at the bottom indicates the zoom level is 80%, the encoding is UTF-8, the line ending is LF, the file type is script, and the cursor is at line 16, column 1.

PRELIMINARY RESULTS

Model Performance Comparison:

Model	MAE (sec)	RMSE (sec)
Linear Regression	384.398	548.9473
Neural Network	381.8115	547.8768

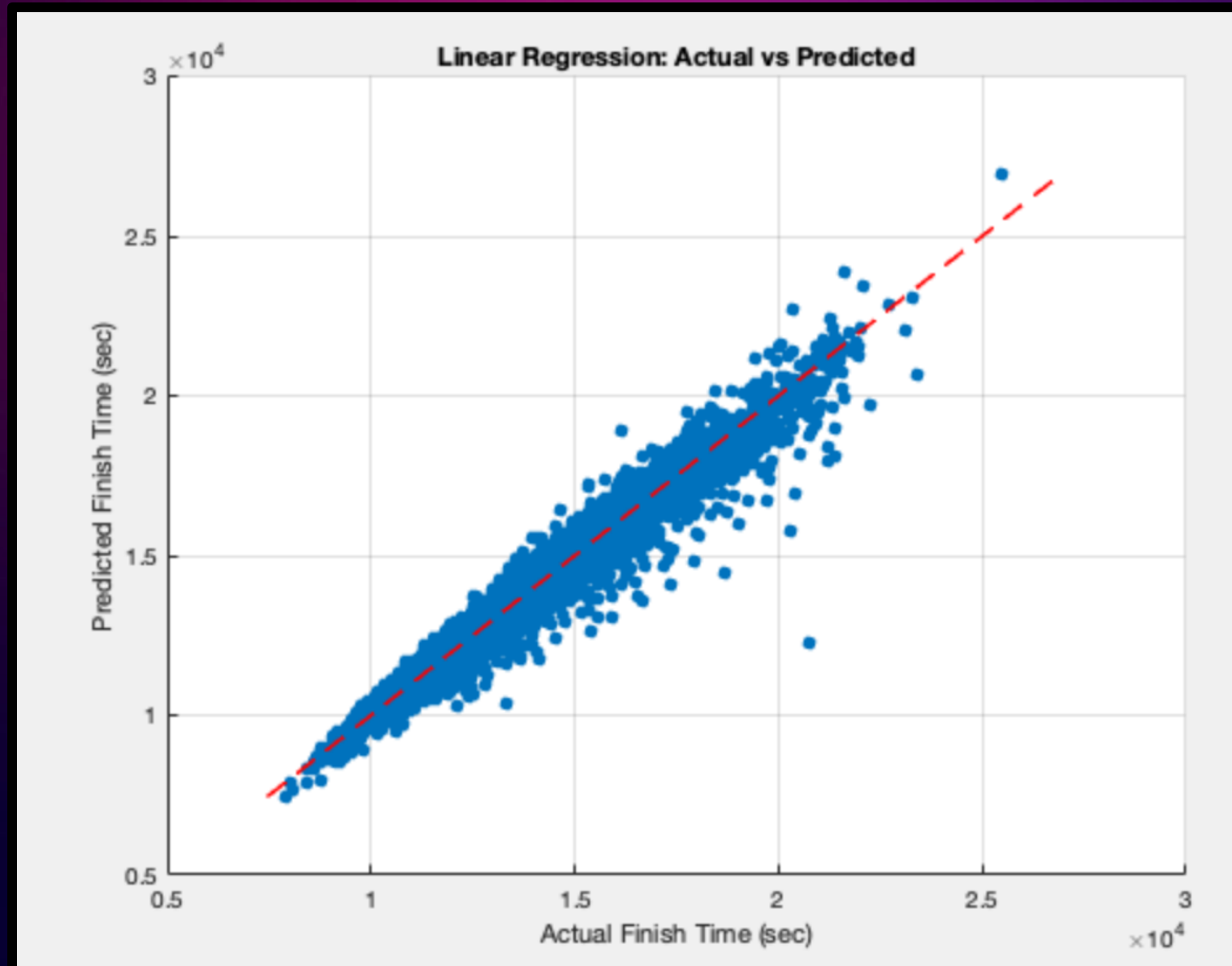
Observations:

- Neural network slightly outperforms linear regression.
- Both models perform similarly.
- Half marathon split times dominates prediction.

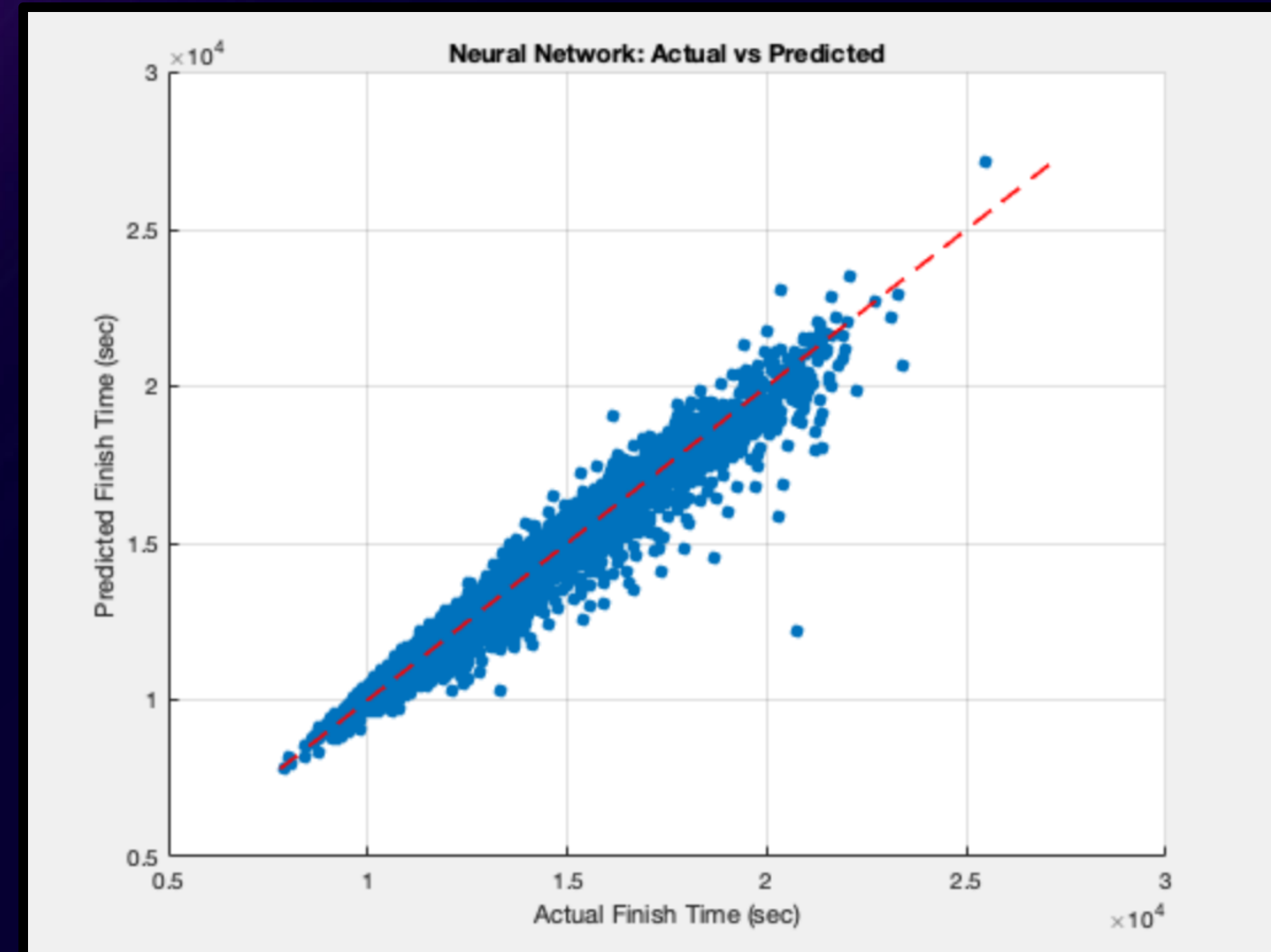
ACTUAL VS. PREDICTED RESULTS

Model Prediction Visualization:

Linear Regression Prediction Plot



Neural Network Prediction Plot



Interpretation:

- Points close to diagonal line = accurate predictions.
- Both models show strong predictive capability.

NEXT STEPS

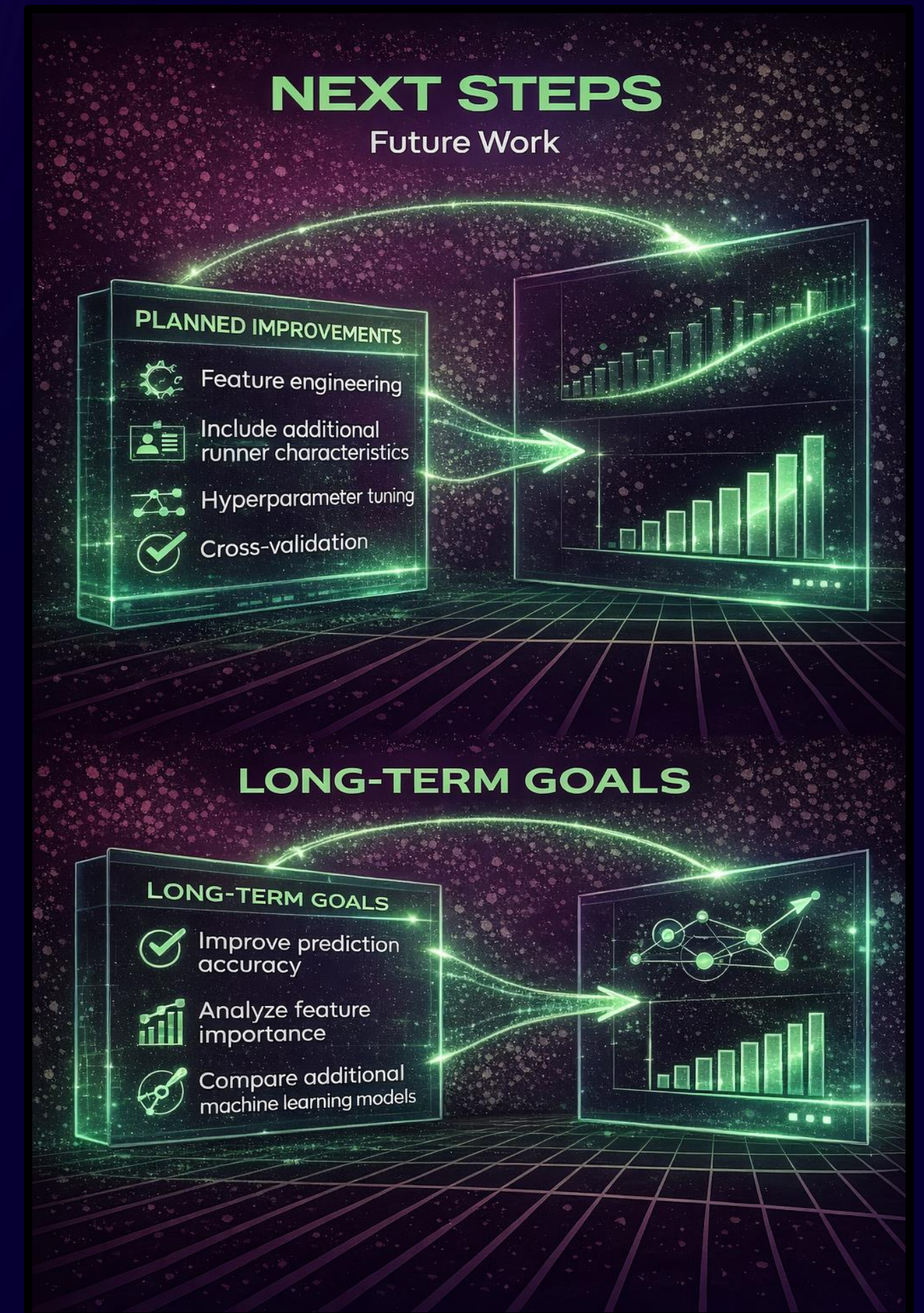
Future Work

Planned improvements:

- Feature engineering
- Include additional runner characteristics
- Hyperparameter tuning
- Cross-validation

Long-term goals:

- Improve prediction accuracy
- Analyze feature importance
- Compare additional machine learning models



Questions?

